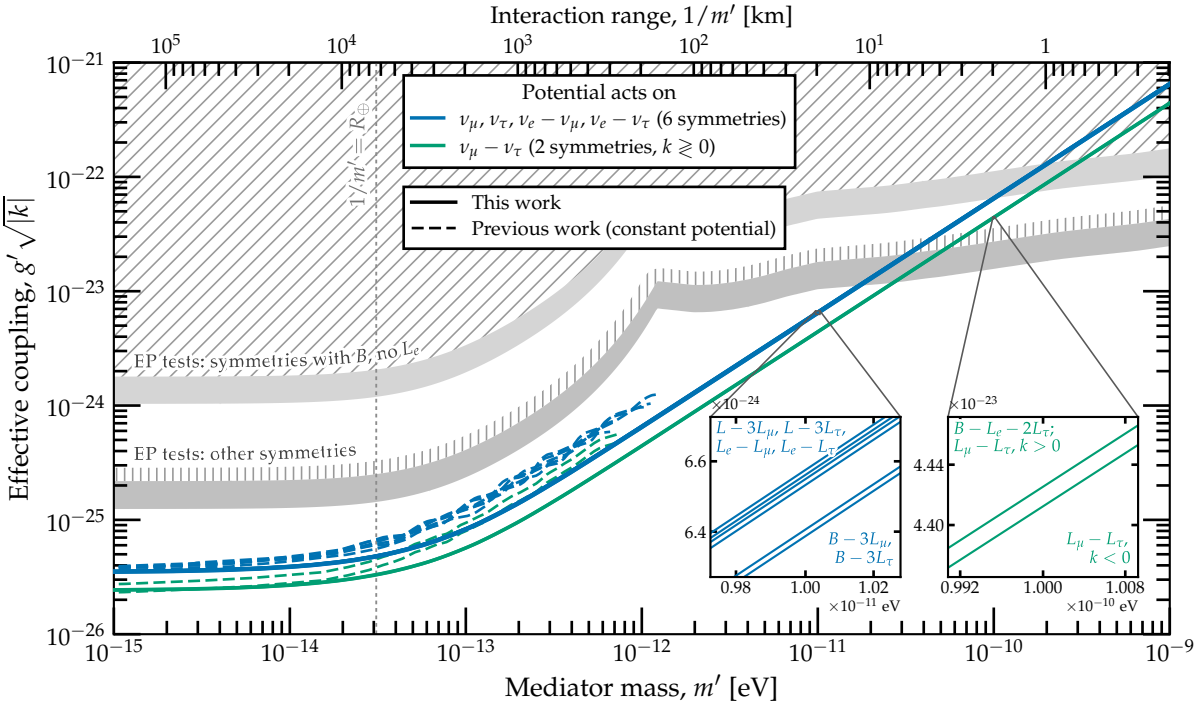

Interaction range, $1/m'$ [km]
10^5
10^4
10^3
10^2
10^1
1
$1/m' \equiv R_\oplus$
Potential acts on
$\nu_\mu, \nu_\tau, \nu_e - \nu_\mu, \nu_e - \nu_\tau$ (6 symmetries)
$\nu_\mu - \nu_\tau$ (2 symmetries, $k \gtrless 0$)
This work
Previous work (constant potential)
EP tests: symmetries with B, no L_e
EP tests: other symmetries
Effective coupling, $g' \sqrt{|k|}$
10^{-21}
10^{-22}
10^{-23}
10^{-24}
10^{-25}
10^{-26}
$\times 10^{-24}$
6.6
6.4
$L - 3L_\mu, L - 3L_\tau,$
$L_e - L_\mu, L_e - L_\tau$
$B - 3L_\mu,$
$B - 3L_\tau$
0.98
1.00
1.02
$\times 10^{-11}$ eV
$\times 10^{-23}$
4.44
4.40
$B - L_e - 2L_\tau;$
$L_\mu - L_\tau, k > 0$
$L_\mu - L_\tau,$
$k < 0$
0.992
1.000
1.008
$\times 10^{-10}$ eV
10^{-15}
10^{-14}
10^{-13}
10^{-12}
10^{-11}
10^{-10}
10^{-9}
Mediator mass, m' [eV]